CHAPTER 9 - Hypothesis testing: One sample

SDS188

2026

1 Types of errors

		Actual situation	
		H_0 is true	H_0 is false
Decision	Do not reject H_0	Confidence $(1-\alpha)$	Type II error β
	Reject H_0	Type I error α	Power $(1-\beta)$

A Type I error can only occur if H_0 is true and 'n Type II error can only occur if H_0 is false. A H_0 can be either true or false. (In an actual situation)

There is thus an inverse relationship between α and β.

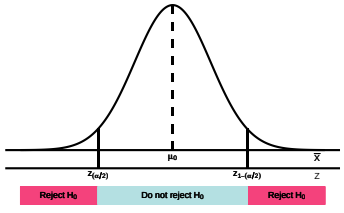
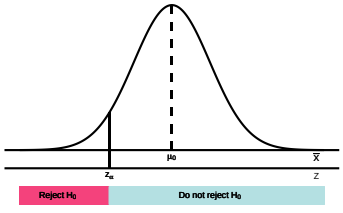
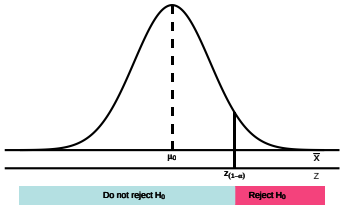
H_0 is true: The probability of rejecting the H_0 is α, and since a Type II error can only occur when H_0 is false, β is basically 0.

H_0 is false: The probability of not rejecting the H_0 is β, and since a Type I error can only occur when H_0 is true, α is basically 0.

2 Hypothesis testing for the population mean

2.1 Critical value approach

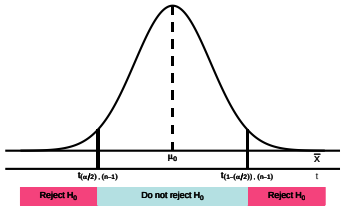
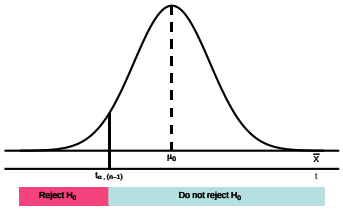
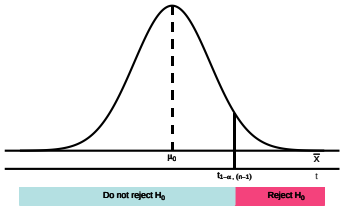
2.1.1 Population standard deviation is known

Step	Two-sided test	Left-sided test	Right-sided test
1	$H_0 : \mu = \mu_0$ vs $H_1 : \mu \neq \mu_0$	$H_0 : \mu \geq \mu_0$ vs $H_1 : \mu < \mu_0$	$H_0 : \mu \leq \mu_0$ vs $H_1 : \mu > \mu_0$
2	α and n		
3	σ is known $\longrightarrow$ Z-test		
4 HP Excel	$z_{\frac{\alpha}{2}}$ and $z_{1-\frac{\alpha}{2}}$ $(\frac{\alpha}{2})$ SHIFT INV M+ 3 and $(1-\frac{\alpha}{2})$ SHIFT INV M+ 3 =NORM.S.INV($\frac{\alpha}{2}$) and =NORM.S.INV($1-\frac{\alpha}{2}$)	z_{α} (α) SHIFT INV M+ 3 =NORM.S.INV(α)	$z_{1-\alpha}$ $(1-\alpha)$ SHIFT INV M+ 3 =NORM.S.INV($1-\alpha$)
5	$z_{stat} = \frac{\bar{x}-\mu_0}{\frac{\sigma}{\sqrt{n}}}$		
6	Reject H_0 if $z_{stat} < z_{\frac{\alpha}{2}}$ or if $z_{stat} > z_{1-\frac{\alpha}{2}}$ =IF(OR((zstat) < (zcritL), (zstat)>(zcritU)), "Reject H0", "Do not reject H0") 	Reject H_0 if $z_{stat} < z_{\alpha}$ =IF((zstat) < (zcrit), "Reject H0", "Do not reject H0") 	Reject H_0 if $z_{stat} > z_{1-\alpha}$ =IF((zstat) > (zcrit), "Reject H0", "Do not reject H0") 

7	Reject H_0 Since $z_{stat} < z_{\frac{\alpha}{2}}$, reject the H_0 and conclude that there is sufficient evidence that $\mu \neq \mu_0$ OR Aangesien $z_{stat} > z_{1-\frac{\alpha}{2}}$, reject the H_0 and conclude that there is sufficient evidence that $\mu \neq \mu_0$ Do not reject H_0 Since z_{stat} is between $z_{\frac{\alpha}{2}}$ and $z_{1-\frac{\alpha}{2}}$, we do not reject the H_0 and conclude that there is insufficient evidence that $\mu \neq \mu_0$	Reject H_0 Since $z_{stat} < z_{\alpha}$, reject the H_0 and conclude that there is sufficient evidence that $\mu < \mu_0$ Do not reject H_0 Since $z_{stat} \geq z_{\alpha}$, we do not reject the H_0 and conclude that there is insufficient evidence that $\mu < \mu_0$	Reject H_0 Since $z_{stat} > z_{1-\alpha}$, reject the H_0 and conclude that there is sufficient evidence that $\mu > \mu_0$ Do not reject H_0 Since $z_{stat} \leq z_{1-\alpha}$, we do not reject the H_0 and conclude that there is insufficient evidence that that $\mu > \mu_0$

2.1.2 Population standard deviation is unknown

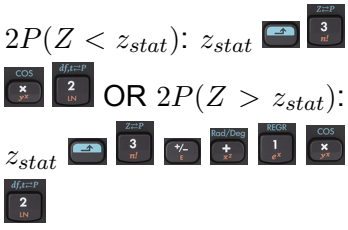
Step	Two-sided test	Left-sided test	Right-sided test
1	$H_0 : \mu = \mu_0$ vs $H_1 : \mu \neq \mu_0$	$H_0 : \mu \geq \mu_0$ vs $H_1 : \mu < \mu_0$	$H_0 : \mu \leq \mu_0$ vs $H_1 : \mu > \mu_0$
2	α and n		
3	σ is unknown $\longrightarrow$ t-test		
4 HP	$t_{\frac{\alpha}{2},n-1}$ and $t_{1-\frac{\alpha}{2},n-1}$ $(n-1)$ $(\frac{\alpha}{2})$ and $(n-1)$ $(1-\frac{\alpha}{2})$	$t_{\alpha,n-1}$ $(n-1)$ (α)	$t_{1-\alpha,n-1}$ $(n-1)$ $(1-\alpha)$

Excel	=T.INV($\frac{\alpha}{2}$,$n-1$) and =T.INV($1-\frac{\alpha}{2}$,$n-1$)	=T.INV(α,$n-1$)	=T.INV($1-\alpha$,$n-1$)
5	$t_{stat} = \frac{\bar{x}-\mu_0}{\frac{s}{\sqrt{n}}}$		
6	Reject H_0 if $t_{stat} < t_{\frac{\alpha}{2},n-1}$ or if $t_{stat} > t_{1-\frac{\alpha}{2},n-1}$ =IF(OR((tstat) < (tcritL), (tstat)>(tcritU)), "Reject H0", "Do not reject H0") 	Reject H_0 if $t_{stat} < t_{\alpha,n-1}$ =IF((tstat) < (tcrit), "Reject H0", "Do not reject H0") 	Reject H_0 if $t_{stat} > t_{1-\alpha,n-1}$ =IF((tstat) > (tcrit), "Reject H0", "Do not reject H0") 
7	Reject H_0 Since $t_{stat} < t_{\frac{\alpha}{2},n-1}$, reject the H_0 and conclude that there is sufficient evidence that $\mu \neq \mu_0$ OR Since $t_{stat} > t_{1-\frac{\alpha}{2},n-1}$, reject the H_0 and conclude that there is sufficient evidence that $\mu \neq \mu_0$	Reject H_0 Since $t_{stat} < t_{\alpha,n-1}$, reject the H_0 and conclude that there is sufficient evidence that $\mu < \mu_0$	Reject H_0 Since $t_{stat} > t_{1-\alpha,n-1}$, reject the H_0 and conclude that there is sufficient evidence that $\mu > \mu_0$

	Do not reject H_0 Since t_{stat} is between $t_{\frac{\alpha}{2},n-1}$ and $t_{1-\frac{\alpha}{2},n-1}$, we do not reject the H_0 and conclude that there is insufficient evidence that $\mu \neq \mu_0$	Do not reject H_0 Since $t_{stat} \geq t_{\alpha,n-1}$, we do not reject the H_0 and conclude that there is insufficient evidence that $\mu < \mu_0$	Do not reject H_0 Since $t_{stat} \leq t_{1-\alpha,n-1}$, we do not reject the H_0 and conclude that there is insufficient evidence that $\mu > \mu_0$

2.2 p-value approach

2.2.1 Population standard deviation is known

Step	Two-sided test	Left-sided test	Right-sided test
1	$H_0 : \mu = \mu_0$ vs $H_1 : \mu \neq \mu_0$	$H_0 : \mu \geq \mu_0$ vs $H_1 : \mu < \mu_0$	$H_0 : \mu \leq \mu_0$ vs $H_1 : \mu > \mu_0$
2	α and n		
3	σ is known $\longrightarrow$ Z-test		
4.1	$z_{stat} = \frac{\bar{x}-\mu_0}{\frac{\sigma}{\sqrt{n}}}$		
4.2	If $z_{stat} < 0$: p-value = $2P(Z < z_{stat})$, if $z_{stat} > 0$: p-value = $2P(Z > z_{stat})$	p-value = $P(Z < z_{stat})$	p-value = $P(Z > z_{stat})$
HP	$2P(Z < z_{stat})$: z_{stat}  OR $2P(Z > z_{stat})$: z_{stat}	z_{stat} 	z_{stat} 

Excel	$2P(Z < z_{stat})$: =2*NORM.S.DIST(z_{stat},TRUE) OR $2P(Z > z_{stat})$: =2*(1-NORM.S.DIST(z_{stat},TRUE))	=NORM.S.DIST(z_{stat},TRUE)	=1-NORM.S.DIST(z_{stat},TRUE)
5	Reject H_0 if p-value $< \alpha$		
6	Reject H_0 Since p-value $< \alpha$, reject the H_0 and conclude that there is sufficient evidence that $\mu \neq \mu_0$ Do not reject H_0 Since p-value $\geq \alpha$, do not reject the H_0 and conclude that there is insufficient evidence that $\mu \neq \mu_0$	Reject H_0 Since p-value $< \alpha$, reject the H_0 and conclude that there is sufficient evidence that $\mu < \mu_0$ Do not reject H_0 Since p-value $\geq \alpha$, do not reject the H_0 and conclude that there is insufficient evidence that $\mu < \mu_0$	Reject H_0 Since p-value $< \alpha$, reject the H_0 and conclude that there is sufficient evidence that $\mu > \mu_0$ Do not reject H_0 Since p-value $\geq \alpha$, do not reject the H_0 and conclude that there is insufficient evidence that $\mu > \mu_0$ nie

2.2.2 Population standard deviation is unknown

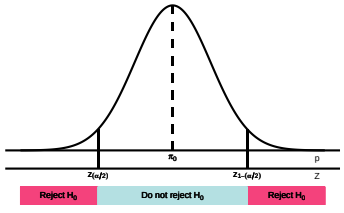
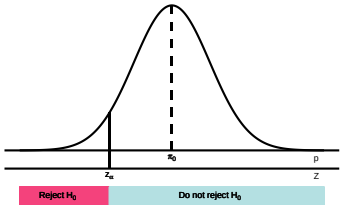
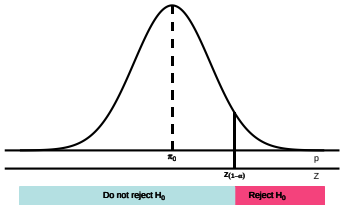
Step	**Two-sided test**	**Left-sided test**	**Right-sided test**
1	$H_0 : \mu = \mu_0$ vs $H_1 : \mu \neq \mu_0$	$H_0 : \mu \geq \mu_0$ vs $H_1 : \mu < \mu_0$	$H_0 : \mu \leq \mu_0$ vs $H_1 : \mu > \mu_0$
2	α and n		
3	σ is unknown $\longrightarrow$ t-test		
4.1	$t_{stat} = \frac{\bar{x}-\mu_0}{\frac{s}{\sqrt{n}}}$		
4.2	If $t_{stat} < 0$: p-value = $2P(t < t_{stat})$, if $t_{stat} > 0$: p-value = $2P(t > t_{stat})$	p-value = $P(t < t_{stat})$	p-value = $P(t > t_{stat})$

HP Excel	$2P(t < t_{stat})$: $(n-1)$ t_{stat} OR $2P(t > t_{stat})$: $(n-1)$ t_{stat} $2P(t < t_{stat})$: =T.DIST.2T(t_{stat},$n-1$)	$(n-1)$ t_{stat} =T.DIST(t_{stat},$n-1$,TRUE)	$(n-1)$ t_{stat} =T.DIST.RT(t_{stat},$n-1$)
5	Reject H_0 if p-value $< \alpha$		
6	Reject H_0 Since p-value $< \alpha$, reject the H_0 and conclude that there is sufficient evidence that $\mu \neq \mu_0$ Do not reject H_0 Since p-value $\geq \alpha$, do not reject the H_0 and conclude that there is insufficient evidence that $\mu \neq \mu_0$	Reject H_0 Since p-value $< \alpha$, reject the H_0 and conclude that there is sufficient evidence that $\mu < \mu_0$ Do not reject H_0 Since p-value $\geq \alpha$, do not reject the H_0 and conclude that there is insufficient evidence to conclude that $\mu < \mu_0$	Reject H_0 Since p-value $< \alpha$, reject the H_0 and conclude that there is sufficient evidence that $\mu > \mu_0$ Do not reject H_0 Since p-value $\geq \alpha$, do not reject the H_0 and conclude that there is insufficient evidence that $\mu > \mu_0$

3 Hypothesis testing for the population proportion

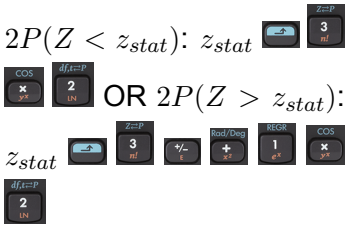
3.1 Critical value approach

Step	Two-sided test	Left-sided test	Right-sided test
1	$H_0 : \pi = \pi_0$ vs $H_1 : \pi \neq \pi_0$	$H_0 : \pi \geq \pi_0$ vs $H_1 : \pi < \pi_0$	$H_0 : \pi \leq \pi_0$ vs $H_1 : \pi > \pi_0$
2	α and n		

3 HP Excel	$z_{\frac{\alpha}{2}}$ and $z_{1-\frac{\alpha}{2}}$ $(\frac{\alpha}{2})$  and $(1-\frac{\alpha}{2})$  =NORM.S.INV($\frac{\alpha}{2}$) and =NORM.S.INV($1-\frac{\alpha}{2}$)	z_{α} (α)  =NORM.S.INV(α)	$z_{1-\alpha}$ $(1-\alpha)$  =NORM.S.INV($1-\alpha$)
4	$z_{stat} = \frac{p-\pi_0}{\sqrt{\frac{\pi_0(1-\pi_0)}{n}}}$		
5	Reject H_0 if $z_{stat} < z_{\frac{\alpha}{2}}$ or if $z_{stat} > z_{1-\frac{\alpha}{2}}$ =IF(OR((zstat) < (zcritL), (zstat)>(zcritU)), "Reject H0", "Do not reject H0") 	Reject H_0 if $z_{stat} < z_{\alpha}$ =IF((zstat) < (zcrit), "Reject H0", "Do not reject H0") 	Reject H_0 if $z_{stat} > z_{1-\alpha}$ =IF((zstat) > (zcrit), "Reject H0", "Do not reject H0") 
6	Reject H_0 Since $z_{stat} < z_{\frac{\alpha}{2}}$, reject the H_0 and conclude that there is sufficient evidence that $\mu \neq \mu_0$ OR Since $z_{stat} > z_{1-\frac{\alpha}{2}}$, reject the H_0 and conclude that there is sufficient evidence that $\mu \neq \mu_0$	Reject H_0 Since $z_{stat} < z_{\alpha}$, reject the H_0 and conclude that there is sufficient evidence that $\mu < \mu_0$	Reject H_0 Since $z_{stat} > z_{1-\alpha}$, reject the H_0 and conclude that there is sufficient evidence that $\mu > \mu_0$

	Do not reject H_0 Since z_{stat} is between $z_{\frac{\alpha}{2}}$ and $z_{1-\frac{\alpha}{2}}$, do not reject the H_0 and conclude that there is insufficient evidence that $\mu \neq \mu_0$	Do not reject H_0 Since $z_{stat} \geq z_{\alpha}$, do not reject the H_0 and conclude that there is insufficient evidence that $\mu < \mu_0$	Do not reject H_0 Since $z_{stat} \leq z_{1-\alpha}$, do not reject the H_0 and conclude that ther is insufficient evidence that $\mu > \mu_0$

3.2 p-value approach

Step	Two-sided test	Left-sided test	Right-sided test
1	$H_0 : \pi = \pi_0$ vs $H_1 : \pi \neq \pi_0$	$H_0 : \pi \geq \pi_0$ vs $H_1 : \pi < \pi_0$	$H_0 : \pi \leq \pi_0$ vs $H_1 : \pi > \pi_0$
2	α and n		
3.1	$z_{stat} = \frac{\pi - \pi_0}{\sqrt{\frac{\pi_0(1-\pi_0)}{n}}}$		
3.2	If $z_{stat} < 0$: p-value = $2P(Z < z_{stat})$, if $z_{stat} > 0$: p-value = $2P(Z > z_{stat})$	p-value = $P(Z < z_{stat})$	p-value = $P(Z > z_{stat})$
HP	$2P(Z < z_{stat})$: z_{stat}  OR $2P(Z > z_{stat})$: z_{stat}	z_{stat} 	z_{stat} 
Excel	$2P(Z < z_{stat})$: =2*NORM.S.DIST(z_{stat},TRUE) OR $2P(Z > z_{stat})$: =2*(1-NORM.S.DIST(z_{stat},TRUE))	=NORM.S.DIST(z_{stat},TRUE)	=1-NORM.S.DIST(z_{stat},TRUE)
4	Reject H_0 if p-value $< \alpha$		

5	Reject H_0 Since p-value $<$ α, reject the H_0 and conclude that there is sufficient evidence that $\pi \neq \pi_0$ Do not reject H_0 Since p-value $\geq \alpha$, do not reject the H_0 and conclude that there is insufficient evidence that $\pi \neq \pi_0$	Reject H_0 Since p-value $<$ α, reject the H_0 and conclude that there is sufficient evidence that $\pi < \pi_0$ Do not reject H_0 Since p-value $\geq \alpha$, do not reject the H_0 and conclude that there is insufficient evidence that $\pi < \pi_0$	Reject H_0 Since p-value $<$ α, reject the H_0 and conclude that there is sufficient evidence that $\pi > \pi_0$ Do not reject H_0 Since p-value $\geq \alpha$, do not reject the H_0 and conclude that there is insufficient evidence that $\pi > \pi_0$

4 Power of the test

4.1 Left-sided test (example in slides)

$$H_0 : \mu = 368$$

$$H_1 : \mu < 368$$

μ	360
σ	15
n	25
α	0.05

Left-sided test $\longrightarrow$ lower critical value

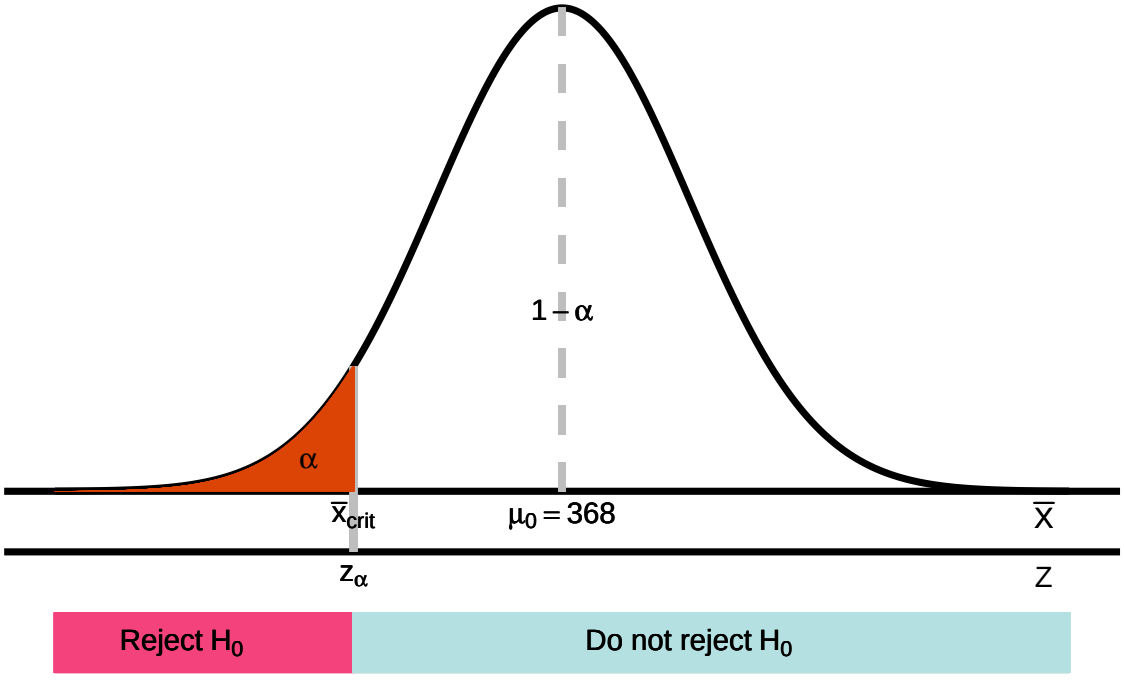
$$z_{crit} = z_\alpha = z_{0.05} = -1.645$$

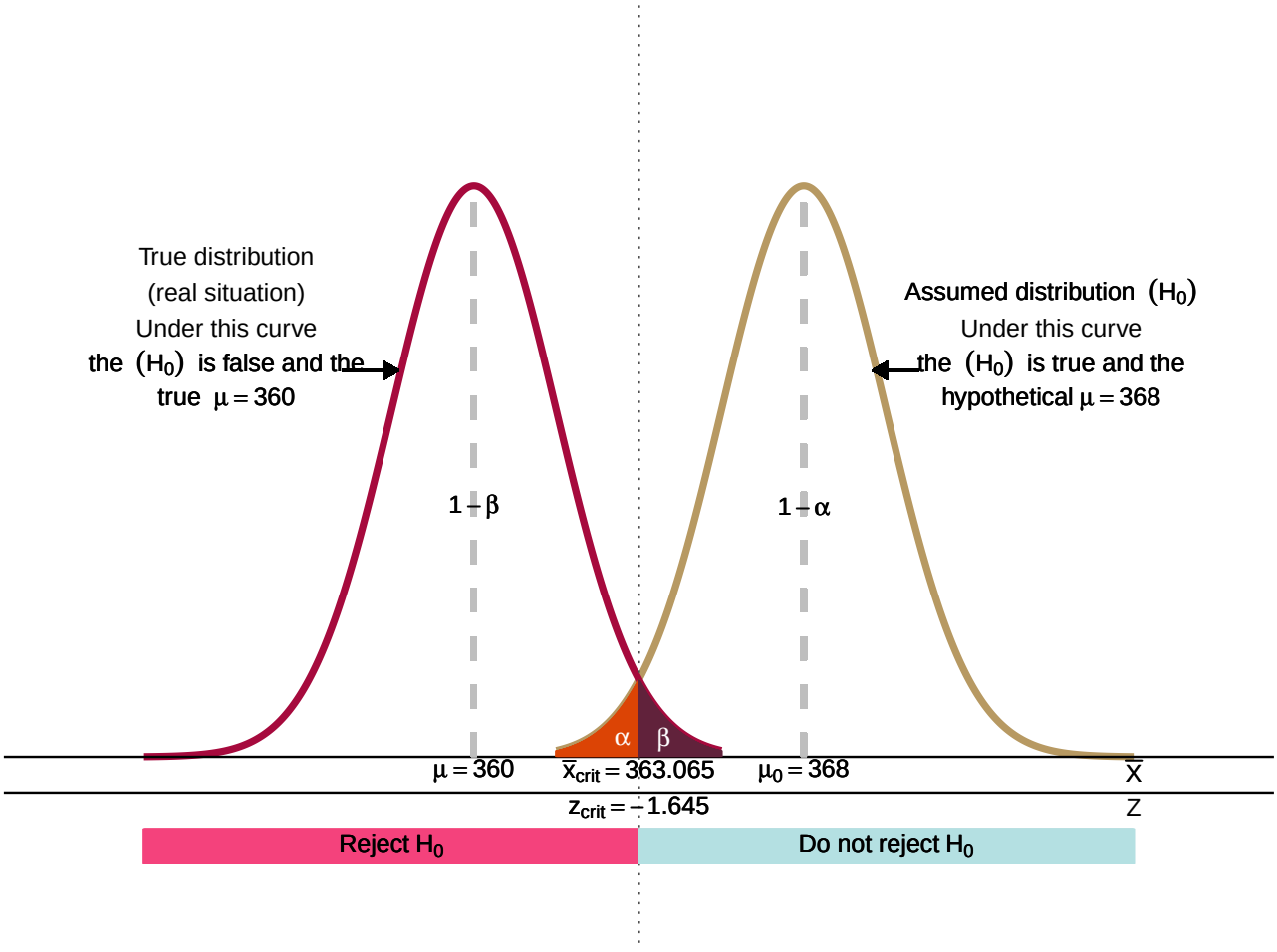
Think of z_{crit} as a standardised value of a sample mean $\bar{x}_{crit}$

$$\begin{aligned} z_{crit} &= \frac{\bar{x}_{crit} - \mu_0}{\frac{\sigma}{\sqrt{n}}} \\ -1.645 &= \frac{\bar{x}_{crit} - 368}{\frac{15}{\sqrt{25}}} \\ \therefore \bar{x}_{crit} &= 368 + (-1.645 \times \frac{15}{\sqrt{25}}) \\ &= 363.065 \end{aligned}$$

Reject H_0 if $z_{stat} < -1.645$

Reject H_0 if $\bar{X} < \bar{x}_{crit} = 363.065$


1 – α
α
$\overline{X}_{crit}$
$\mu_0 = 368$
$\overline{X}$
z_α
Z
Reject H_0
Do not reject H_0



$$\alpha = P(\text{Reject } H_0 \mid H_0 \text{ is true})$$

$$1 - \alpha = P(\text{Do not reject } H_0 \mid H_0 \text{ is true})$$

$$\beta = P(\text{Do not reject } H_0 \mid H_0 \text{ is false})$$

$$1 - \beta = P(\text{Reject } H_0 \mid H_0 \text{ is false})$$

$$
\begin{aligned}
\beta &= P(\text{Do not reject } H_0 | H_0 \text{ is false}) \\
&= P(\bar{X} > \bar{x}_{crit} | \mu = 360) \\
&= P(Z > \frac{\bar{x}_{crit} - \mu}{\frac{\sigma}{\sqrt{n}}} \\
&= P(Z > \frac{363.065 - 360}{\frac{15}{\sqrt{25}}} \\
&= P(Z > 1.0217) \\
&= 1 - P(Z < 1.0217) \\
&= 1 - 0.8461 \\
&= 0.1539
\end{aligned}
$$

$$1 - \beta = 1 - 0.1539 = 0.8461$$

4.2 Right-sided test

$$H_0 : \mu = 368$$

$$H_1 : \mu > 368$$

μ	380
σ	15
n	25
α	0.05

Right-sided test $\longrightarrow$ upper critical value

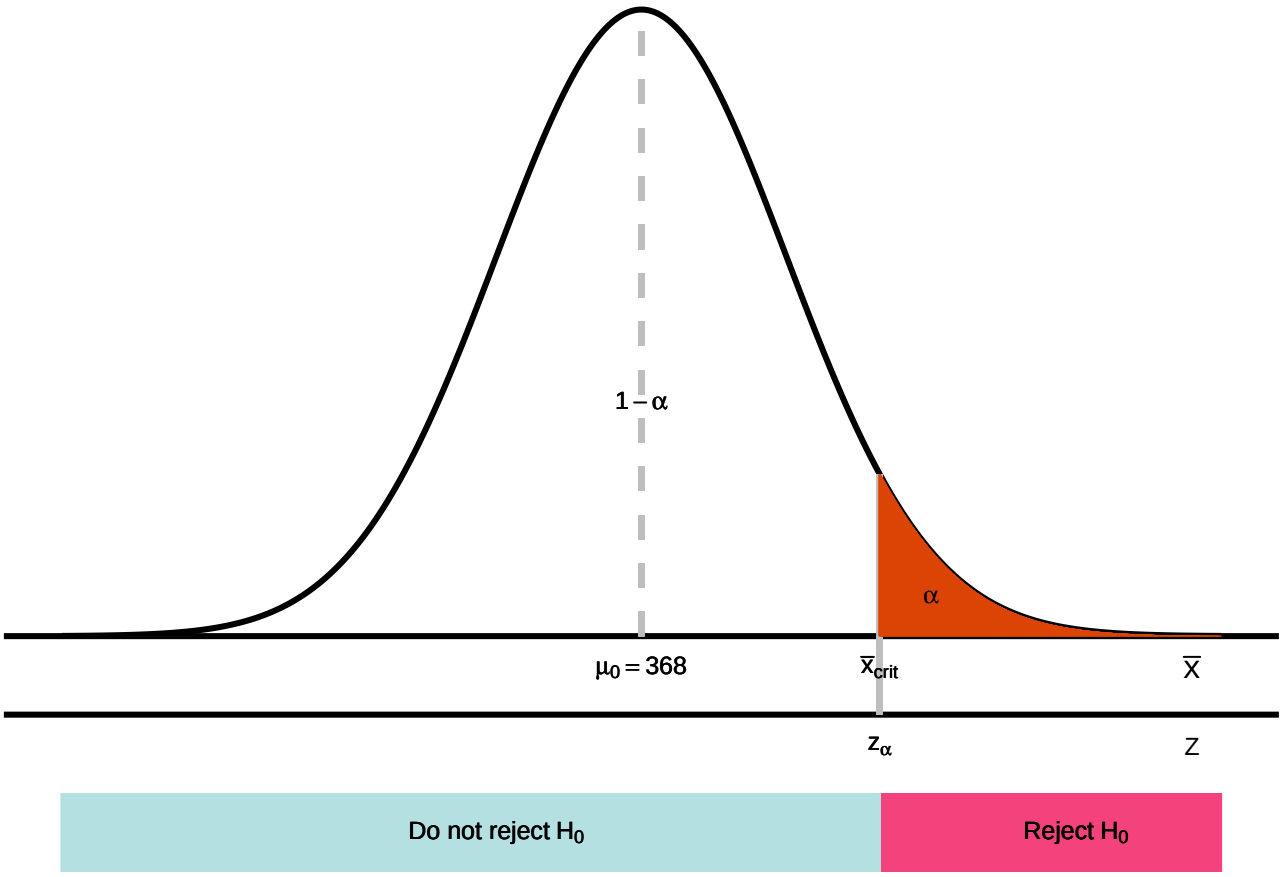
$$z_{crit} = z_{1-\alpha} = z_{0.95} = 1.645$$

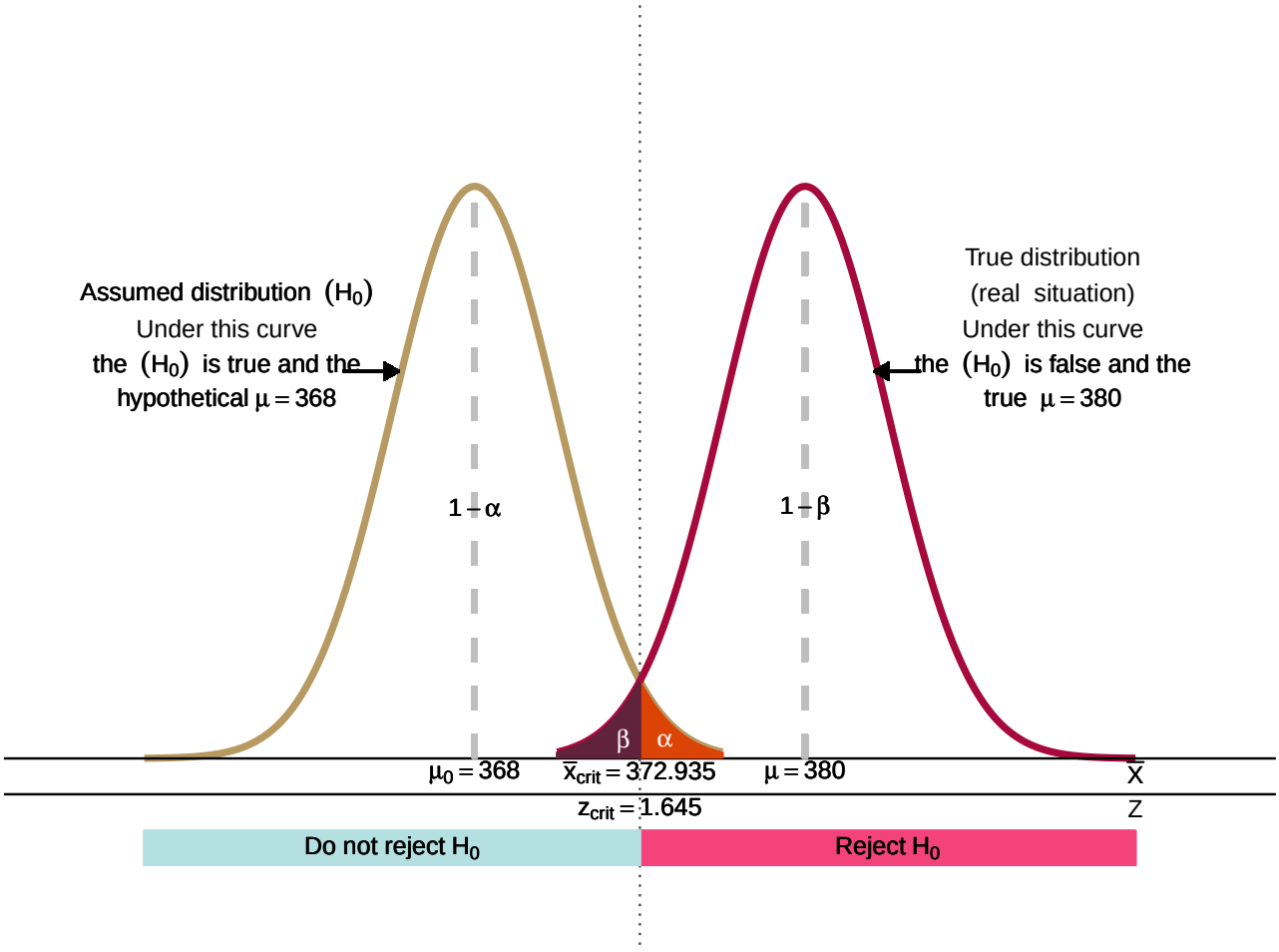
Think of z_{crit} as a standardised value of a sample mean $\bar{x}_{crit}$

$$\begin{aligned} z_{crit} &= \frac{\bar{x}_{crit} - \mu_0}{\frac{\sigma}{\sqrt{n}}} \\ 1.645 &= \frac{\bar{x}_{crit} - 368}{\frac{15}{\sqrt{25}}} \\ \therefore \bar{x}_{crit} &= 368 + (1.645 \times \frac{15}{\sqrt{25}}) \\ &= 372.935 \end{aligned}$$

Reject H_0 if $z_{stat} > 1.645$

Reject H_0 if $\bar{X} > \bar{x}_{crit} = 372.935$


1 – α
α
μ0 = 368
x̄crit
X̄
zα
Z
Do not reject H0
Reject H0



$$\alpha = P(\text{Reject } H_0 \mid H_0 \text{ is true})$$

$$1 - \alpha = P(\text{Do not reject } H_0 \mid H_0 \text{ is true})$$

$$\beta = P(\text{Do not reject } H_0 \mid H_0 \text{ is false})$$

$$1 - \beta = P(\text{Reject } H_0 \mid H_0 \text{ is false})$$

$$
\begin{aligned}
\beta &= P(\text{Do not reject } H_0 | H_0 \text{ is false}) \\
&= P(\bar{X} < \bar{x}_{crit} | \mu = 380) \\
&= P(Z < \frac{\bar{x}_{crit} - \mu}{\frac{\sigma}{\sqrt{n}}} \\
&= P(Z < \frac{372.935 - 380}{\frac{15}{\sqrt{25}}} \\
&= P(Z < -2.3551) \\
&= 0.00926
\end{aligned}
$$

$$
1 - \beta = 1 - 0.00926 = 0.9907
$$

4.3 General steps

1. Calculate critical value
2. Think of the critical value as a standardised value of sample value from the sampling distribution of the statistic.
    - Determine that statistic $\hat{\theta}_{crit} = \mu_{\hat{\theta}} + (\text{critical value} \times \sigma_{\hat{\theta}})$
3. Given the true parameter (H_0 is false), calculate $\beta = P(\text{Do not reject } H_0 | H_0 \text{ is false})$ and/or $1 - \beta = P(\text{Reject } H_0 | H_0 \text{ is false})$

5 Questions

5.1 Question 1

During a period of a month, a random sample of 27 approved life insurance policies is selected. The total processing times (in days) are given as follows:

72	19	16	64	28	28
31	90	64	54	33	56
22	18	50	48	27	17
17	91	83	63	50	51
69	16	17			

In the past, the mean processing time was 49 days and it is being tested that it could be more than 49 days. Calculate the probability of a Type II error if the significance level is 1% and the actual mean is 52.

5.2 Question 2

You are a restaurant manager for a fast food franchise. In the previous month, the average waiting time at the drive-thru window for branches in your area was 3.6 minutes. The waiting time is measured from the time a client places an order until the time receives the order. A random sample of 64 orders are selected where the average time is 3.83 minutes and the sample standard deviation is 0.8 minutes. Given a significance level of 10%, is there any evidence to suggest that the population mean waiting time differs from 3.6 minutes? Use the p-value approach.

5.3 Question 3

A financial analyst is investigating the average credit card debt incurred by senior students from German universities. She determines that the average debt incurred by a senior student is €4600. Ac-

cording to the student council report from Dortmund University, the average debt incurred is less than €4600. A random sample of 50 senior students is selected of which the average debt is €4201. The population standard deviation is assumed to be €1500.

1. For a significance level of 0.01, is there any evidence to suggest that the average debt differs from €4600. Use the critical value approach.
2. Calculate the p-value and use it to reach a conclusion at a 0.01 significance level.
3. Construct a 99% confidence interval for the population average amount of debt incurred.
4. Compare the conclusions.

5.4 Question 4

A company that produces chocolate bars wants to determine if the average weight of a chocolate bar is not more than 170g. A random sample of 50 chocolate bars are selected. The sample mean is 172g and the sample standard deviation is 0.6g. For a significance level of 1%, is there any evidence to suggest that the average weight of the chocolate bars is more than 170g? Use the critical value approach.